

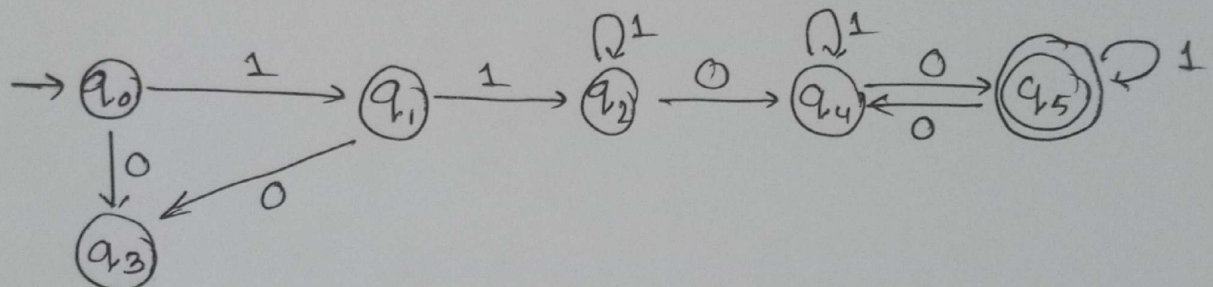
1(a)

Computability theory: Two states of an Automata are said to be compatible if there is no input string that can distinguish them.

1(b)

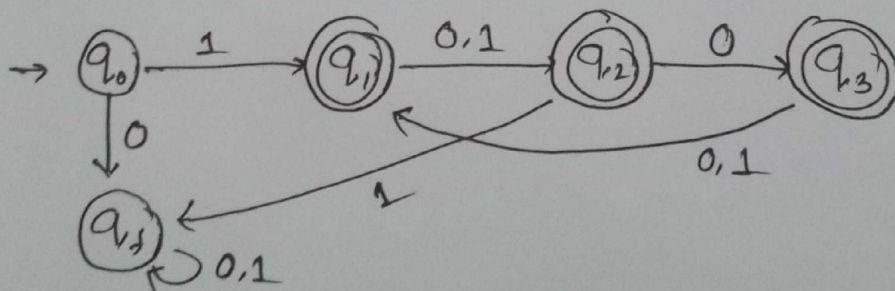
i) $\{w \mid w \text{ begins with } 11 \text{ and has even number of } 0\text{'s}\}$

$$L(w) = \{1100, 11100001, 110000, \dots\}$$



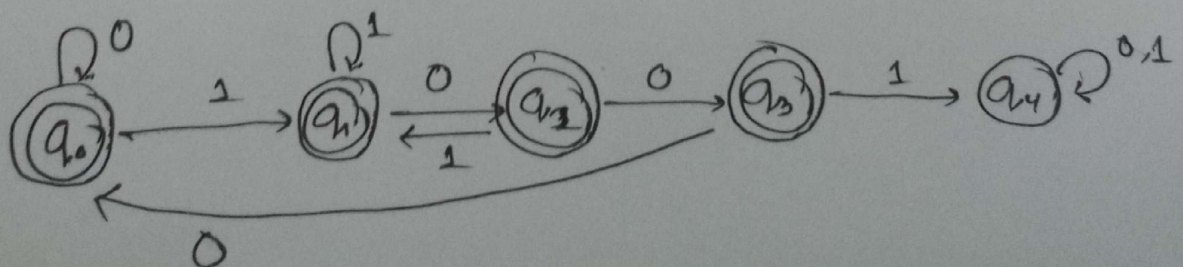
ii) $\{w \mid w \text{ begins with } 1 \text{ and every third symbol } 0\}$

$$L(w) = \{110, 100110, 110110, \dots\}$$



iii) $\{w \mid w \text{ does not contain substring } 1001\}$

$$L(w) = \{0, 00, 1101, 100, \dots\}$$



1(c)

i) begin with 11 and has even number of 0

$$\Rightarrow 11(1^*(01^*01^*)^*)$$

ii) begin with a 1 and every third symbol 0

$$\Rightarrow 1((0/1)0(0/1))^*$$

iii) does not contain the substring 1001

$$(0/1)^* - (0/1)^*1001(0/1)^*$$

2(a)

accepted string

→ bbabab

→ babab

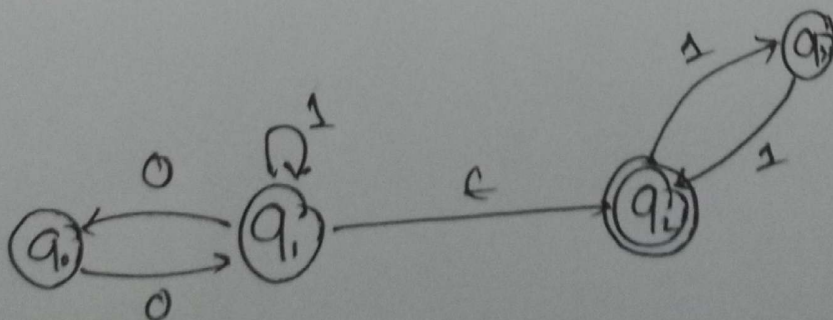
not accepted string

→ bab

→ bbbba

(OR)

$$(00 \cup 1)^*(11)^*$$



2(b)

i) $(11 \cup 00 \cup 10 \cup 01)^*$

⇒ The language consists of all strings having even number of length.

ii) $(00)^* 1 (11)^*$

⇒ The language consists of all binary strings that contain exactly one 1 positioned between an even number of 0's (possibly none) and an even number of 1's (possibly none).

iii) $(00 \cup 1)^*$

⇒ the language consists of all binary strings in which 0's always appear in pairs and 1's may appear anywhere.

Ans to the question no 2(c)

Proof:

The class of regular languages is closed under the union operation:

$N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognise A_1 ,

$N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognise A_2 ,

let construct $N = (Q, \Sigma, \delta, q_0, F)$ to recognise $A_1 \cup A_2$

$$\textcircled{1} Q = \{q_0\} \cup Q_1 \cup Q_2$$

The states of N are all the states of N_1 and N_2 with the addition of a new state q_0 .

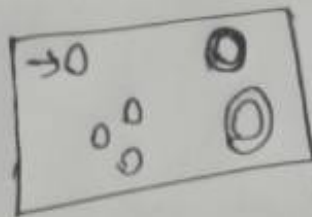
② The state q_0 is the start of N

③ The set of accept states $F = F_1 \cup F_2$. The accept states of N are all the accept states of N_1 and N_2 . That way, N accepts if either N_1 accepts or N_2 accepts.

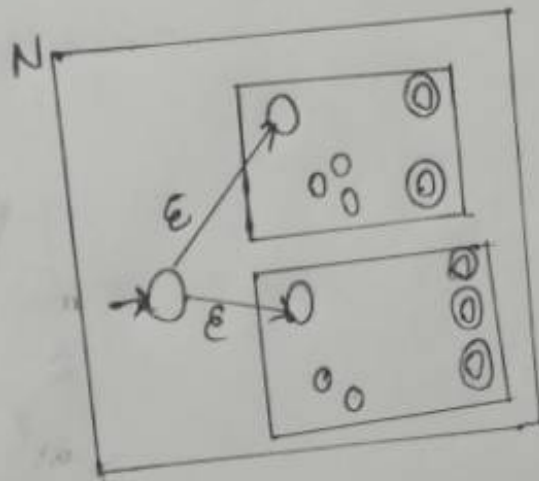
④ Define δ so that for any $q \in Q$ and any $E \in \Sigma$,

$$\delta(q, a) = \begin{cases} \delta_1(q, a) & q \in Q_1 \\ \delta_2(q, a) & q \in Q_2 \\ \{q_1, q_2\} & q = q_0 \text{ and } a = \epsilon \\ \emptyset & q = q_0 \text{ and } a \neq \epsilon \end{cases}$$

N_1



N_2



3(a)

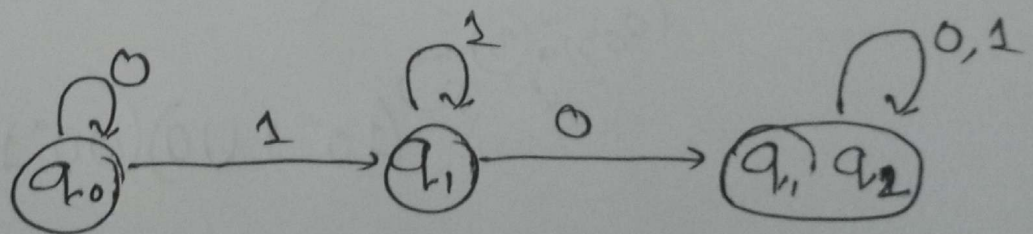
NFA Transition table

State	0	1
q_0	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_1, q_2\}$

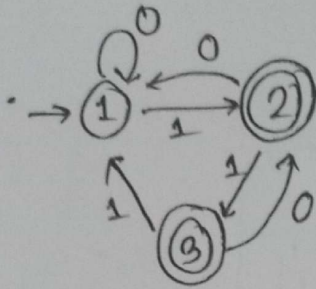
DFA Transition table

State	0	1
q_0	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
q_1, q_2	$\{q_1, q_2\}$	$\{q_1, q_2\}$

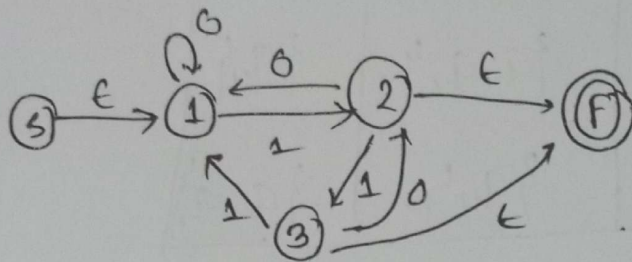
Converted DFA diagram:



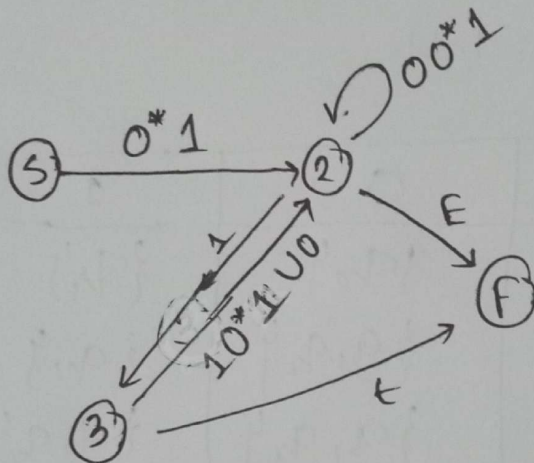
3(b)



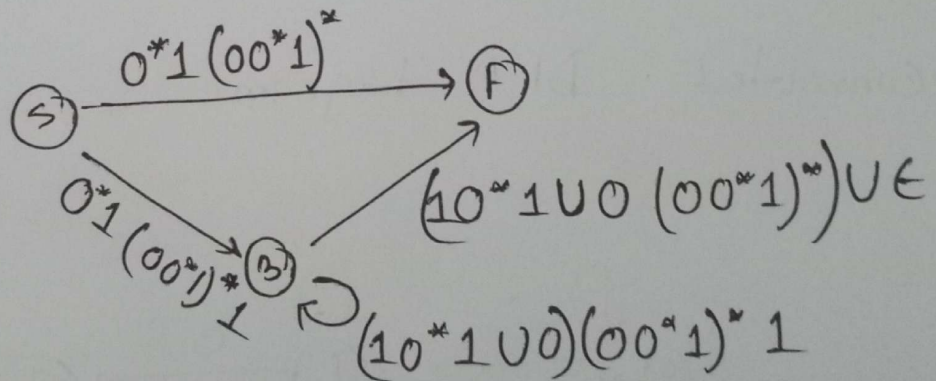
Step 1



Step 2



Step 3



Step 4

